SEED: Self-Evolving On-Policy Distillation

Stage 1: Hindsight Skill SFT

1 Offline trajectory collection

q_1
q_2
...
q_n

Sampled tasks

Base Model

Environment

Interact with environment

2 Hindsight skill annotation

Observations	$o_1, o_2, o_3 \dots o_T$
Actions	$a_1, a_2, a_3 \dots a_T$
Rewards	$r_1, r_2, r_3 \dots r_T$
Outcome	success/failure

Full trajectory records

External Analyzer

Extract Skills

3 Supervised fine-tuning

- reusable strategy
- failure correction
- ...

Hindsight skills

SFT Data

SFT Model

Initialize RL policy

Stage 2: Self-Evolving On-Policy Distillation

Self-Evolving Loop

Actor

Shared Model

Analyzer

Trajectories Group

$\{\tau_q^{(1)}, \dots, \tau_q^{(N)}\}$

$\tau_q^{(1)}$ ... $\tau_q^{(N)}$

Completed Trajectories

Group-Relative Advantage

$$A_{q,n}^{\mathrm{rl}} = \frac{R\left(\tau_q^{(n)}\right) - \mu_q}{\sigma_q}$$

RL Loss: $\mathcal{L}_{RL}$

SEED Loss:
$\mathcal{L}_{SEED} = \mathcal{L}_{RL} + \mathcal{L}_{OPD}$

Updated Policy

Hindsight Skills

$\{s_q^{(1)}, \dots, s_q^{(N)}\}$

$s_q^{(1)}$ ... $s_q^{(N)}$

Insert hindsight skill to original context

Skill-Augmented Context

$\tilde{h}_{q,n} = H(h_{q,n}, s_q^{(n)})$

$h_{q,n}$ Original Context

Insert $s_q^{(n)}$

Same sampled actions with different contexts

Re-score on sampled actions

Teacher: $\log \pi_\theta(a|\tilde{h}_{q,n})$

Student: $\log \pi_\theta(a|h_{q,n})$

OPD Loss: $\mathcal{L}_{OPD}$